

KOGI PLATFORM

kogi-bank

Banking, Finance & Capital System

System Design Document

v1.0 · Kogi Platform · March 2026

KWLT · kogi-bank · KFIN · KESC · KLGR

Banking for independent workers, autonomous organizations, collectives, cooperatives, and independent teams — the financial operating layer of the Kogi Platform.

Kogi Team · Independent Worker Operating System · Confidential

Table of Contents

1. OVERVIEW

1.1 Purpose & Vision

kogi-bank is the financial operating layer of the Kogi Platform — the centralized banking, accounting, payment, escrow, investment, and capital management system for independent workers, autonomous organizations, collectives, cooperatives, and independent teams. It is the platform module referenced as KWLT (Kogi Wallet) in earlier design documents, now formalized as kogi-bank to reflect its full scope: it is not merely a wallet but a complete financial operating infrastructure.

Where kogi-office organizes work, kogi-community connects people, and kogi-marketplace enables trade, kogi-bank ensures that all financial flows — from a single invoice to a multi-party equity round — are governed, auditable, and intelligently managed. Every transaction, every account balance, every escrow lock, and every campaign contribution flows through kogi-bank.

Design Principle: Accounts are stores of capital, resources, liquidity, and equity. Wallets are transaction points — the active interfaces through which financial entities move. Every financial event in the platform is a first-class Portfolio Item in the Portfolio System, giving every financial entity the same governance, versioning, and analytics infrastructure as any other platform component.

1.2 Core Design Distinctions: Accounts vs. Wallets

A foundational architectural distinction governs the entire kogi-bank system:

Concept	Description
Account	A store of capital, resources, liquidity, or equity. Accounts hold balances — they are the authoritative record of what is owned or owed. Think: checking account, investment account, escrow reserve, equity stake register.
Wallet	A transaction point — the active interface through which financial items move. Wallets are operational surfaces for sending, receiving, and routing. A wallet links to one or more accounts and executes transactions on their behalf. Think: spending wallet, payout wallet, campaign wallet.
Ledger	The immutable, append-only record of every financial event. The ledger is the source of truth for all account balances — balances are always computed from the ledger, never stored independently.
Journal	A classified grouping of ledger entries by type — income, expense, asset, liability, equity. Journals are the accounting layer above raw ledger entries.

1.3 Position in the Kogi Platform

Layer	System	Relationship to kogi-bank
Upstream	kogi-marketplace / kogi-exchange	All deal completions, escrow operations, and campaign contributions route through kogi-bank
Upstream	kogi-gig	Worker scheduling and contract completions trigger payroll, invoicing, and payment events
Upstream	kogi-portfolio (Portfolio System)	Budget allocations and resource spend recorded via ResourceAllocation (Budget kind)
Downstream	kogi-engine (RiskEngine)	Financial signals (cashFlow, portfolio health) consumed for risk scoring
Downstream	kogi-engine (AnalyticsEngine)	Financial StreamEvents feed the analytics and recommendation pipelines
Peer	kogi-community / kogi-organizations	Cooperative and collective banking, shared funds, campaign disbursements
Peer	kogi-marketplace / kogi-exchange	Escrow lock/release, deal settlement, investment disbursement
External	Stripe, Plaid, PayPal, ACH, Crypto bridges	Payment rails, bank account linking, on-chain settlement

1.4 Module Codes & Services

Module	Code	Service	Language
Bank Core	KBNK	kogi-bank	Go (service) · Rust (BankSystem core)
Wallet	KWLT	kogi-wallet	Go (service)
Ledger	KLGR	kogi-ledger	Go (service) · PostgreSQL (append-only)
Escrow	KESC	kogi-escrow	Go (service) · Rust
Accounts	KACC	kogi-accounts	Go (service)
Campaigns	KCMP	kogi-campaigns	Go (service)
Invoicing	KINV	kogi-invoices	Go (service)
Tax & Accounting	KTAX	kogi-tax	Go (service) · Scala (pipelines)

2. ACCOUNT SYSTEM

2.1 Concept

Accounts are the foundational stores of financial value in kogi-bank. An account represents a defined pool of capital, resources, liquidity, or equity belonging to a specific entity — an individual worker, an organization, a collective, a cooperative, or an independent team. Every account has an owner, a currency or unit denomination, a type classification, and a balance computed from its ledger entries.

Accounts are Portfolio Items at the system level — they carry the full ComponentMetadata structure (id, owners, tags, policy_ids, vector_clock, version) and are governed by the Portfolio System's PermissionTier hierarchy and PolicyEngine. This means an account's full audit trail, governance policies, and access controls are native to the same infrastructure as all other platform entities.

2.2 Account Types

2.2.1 Personal Accounts (Individual Worker)

Account Type	Description	Currency/Unit	Primary Use
Personal Checking	Day-to-day spending and income reception	Fiat (USD, EUR, etc.)	Receive client payments, pay expenses
Personal Savings	Reserved capital for specific goals or emergencies	Fiat	Emergency fund, goal-based saving
Operations Account	Dedicated business-expense account for gig/freelance activity	Fiat	Software subscriptions, equipment, tools
Investment Account	Long-term capital deployment in equity, funds, or portfolio assets	Fiat + equity units	Equity stakes, platform investments
Tax Reserve Account	Automated set-aside for self-employment tax obligations	Fiat	Quarterly tax payments, year-end filing
Retirement Account	Long-term personal retirement savings (portable benefit)	Fiat + fund units	IRA-equivalent, SEP, solo 401(k) analogues
Equity Account	Record of equity positions in external organizations or portfolio projects	Equity units / shares	Cap table holdings, vesting schedules

Resource Account	Non-monetary resource pools (compute credits, license units, hours)	Custom units	Platform compute, license pools, time credits
Trading Account	Active trading and exchange activity	Fiat + tokens	kogi-exchange activity, market orders
Crypto/Token Account	Digital asset and token holdings	Tokens / coins	Web3 assets, platform tokens, NFTs

2.2.2 Organizational Accounts (Autonomous Org, Collective, Cooperative, Team)

When an organization, collective, cooperative, or team is created in kogi-organizations, kogi-bank provisions a corresponding set of organizational accounts. These accounts are owned by the organization entity and governed by its member permissions and governance policies.

Account Type	Description	Entity Types	Notes
Operating Account	Primary revenue and expense account for the organization	Org, Collective, Cooperative, Team	Multi-signatory controls for large withdrawals
Payroll Account	Funds designated for member/worker compensation	Org, Cooperative, Team	Automated distribution per payroll schedule
Project Account	Per-project budget pool; linked to a specific Portfolio Project	All	Budget ring-fenced per project; spend tracked in Portfolio System
Reserve Account	Organizational reserves and contingency capital	Org, Cooperative	Governed by reserve policy; minimum balance enforced
Investment Account	Collective investment capital deployed in portfolio assets or external opportunities	All	Multi-party approval required for deployment
Equity Pool	Shared equity register for member equity distribution	Cooperative, DAO	Records vesting, cliff dates, distribution events
Campaign Account	Dedicated account for an active fundraising or resource campaign	All	Funds held until campaign resolution; auto-disbursed on success
Escrow Reserve	Collective escrow for multi-party deals	All	Funds released per approval workflow
Tax Account	Shared tax reserve for organizational obligations	Org, Cooperative	Automated allocation from operating revenue

Mutual Aid Fund	Peer-support pool for member emergencies or collective needs	Collective, Cooperative	Governed by mutual aid policy
-----------------	--	-------------------------	-------------------------------

2.3 Account Data Model

Field	Description
account.id	Platform-unique UUID
account.type	checking savings operations investment tax_reserve retirement equity resource trading crypto payroll project reserve equity_pool campaign escrow_reserve tax mutual_aid
account.owner_id	Owner entity UUID — user, organization, collective, cooperative, or team
account.owner_type	user organization collective cooperative team
account.name	Display name of the account
account.currency	ISO 4217 currency code (USD, EUR, GBP) or custom unit code
account.denomination	fiat equity_units tokens resource_units custom
account.balance	Computed: sum of all ledger credits minus debits for this account
account.available_balance	Balance minus any pending holds, escrow locks, or reserved amounts
account.pending_balance	Amount currently in unconfirmed or in-flight transactions
account.status	active frozen suspended closed archived
account.linked_external	External account link IDs (via Plaid, Stripe, etc.) if connected
account.linked_wallets[]	Wallet IDs that can transact against this account
account.policy_ids[]	Governance policy IDs: withdrawal limits, multi-sig rules, reserve minimums
account.portfolio_component_id	Linked Portfolio Component ID — account is itself a Portfolio Item
account.metadata	Full ComponentMetadata: id, owners, tags, vector_clock, version, created_at, updated_at

2.4 Account Lifecycle

State	Description	Transitions
Provisioned	Account created but not yet activated. Zero balance.	→ Active (deposit or verification)
Active	Fully operational. Transactions permitted.	→ Frozen, → Suspended, → Closed

Frozen	Temporarily locked. No outbound transactions. Deposits allowed.	→ Active (unfreeze), → Suspended
Suspended	Compliance or policy hold. No transactions.	→ Active (resolution), → Closed
Closed	Account permanently closed. Balance must be zero or transferred.	→ Archived
Archived	Historical record only. No transactions.	(Terminal)

2.5 Multi-Signatory & Approval Controls

Organizational and high-value accounts support configurable multi-signatory controls, enforced through the Portfolio System's ApprovalWorkflow and PolicyEngine:

- Single-sig accounts — standard individual accounts; one authorized user can transact
- 2-of-N multi-sig — two approvers required from a defined set of N authorized signatories
- M-of-N multi-sig — M approvers from N required; configurable per account and per transaction threshold
- Time-locked approvals — transactions above a threshold require a time delay before execution (cooling-off window)
- DAO governance approval — organizational accounts linked to on-chain or platform-native voting for large disbursements
- Delegated authority — specific signatories can be granted time-limited authority to approve up to a specified amount

3. WALLET SYSTEM

3.1 Concept

Wallets are the active transaction surfaces of kogi-bank. While accounts store value, wallets move it. A wallet is a named, purpose-defined transaction interface that routes financial operations to and from one or more underlying accounts. Users interact with wallets daily; accounts are the deeper financial record layer.

Every user and organization has a primary wallet, but can configure multiple purpose-specific wallets — a personal spending wallet, a gig income wallet, a marketplace wallet for buying and selling, and a campaign wallet for fundraising activity. Each wallet has a defined flow direction, linked accounts, transaction rules, and an activity feed.

3.2 Wallet Types

Wallet Type	Description	Typical Linked Accounts	Primary Use
Personal Wallet	Day-to-day financial activity	Personal Checking, Savings	Receive income, pay expenses, transfer funds
Operations Wallet	Business expense management for gig activity	Operations Account	Subscriptions, tools, equipment, platform fees
Marketplace Wallet	Buying and selling in kogi-marketplace	Checking, Investment	Purchase assets, receive sales proceeds
Exchange Wallet	Active trading and deal execution	Trading Account, Crypto	Exchange orders, deal escrow, bid funding
Investment Wallet	Deploying capital in equity, campaigns, and instruments	Investment Account, Equity Account	Back campaigns, invest in rounds, buy equity
Payroll Wallet	Receive compensation from contracts and gigs	Personal Checking	Gig payouts, contract milestone receipts
Tax Wallet	Automated tax savings and payments	Tax Reserve Account	Estimated quarterly taxes, annual filing
Campaign Wallet	Manage and disburse campaign funds	Campaign Account	Launch campaigns, track contributions, disburse
Org Wallet	Organizational operational wallet	Operating Account, Payroll Account	Org-wide spend, payroll, vendor payments
Collective Wallet	Shared wallet for collective/cooperative member activity	Mutual Aid Fund, Reserve Account	Collective purchases, shared resources, mutual aid
Escrow Wallet	Temporary custody of deal funds	Escrow Reserve Account	Deal-in-progress fund custody

Crypto Wallet	Digital asset transactions	Crypto/Token Account	On-chain transfers, token purchases, DeFi activity
---------------	----------------------------	----------------------	--

3.3 Wallet Data Model

Field	Description
wallet.id	Platform-unique UUID
wallet.type	personal operations marketplace exchange investment payroll tax campaign org collective escrow crypto
wallet.owner_id	Owner entity UUID
wallet.name	Display name
wallet.linked_accounts[]	Account IDs this wallet can transact against
wallet.primary_account_id	Default account for this wallet's transactions
wallet.balance_display	Computed aggregate of linked account balances for display
wallet.status	active paused frozen closed
wallet.transaction_rules	JSONB: spending limits, auto-save rules, auto-tax-split rules
wallet.auto_split[]	Automatic percentage routing to linked accounts on incoming funds
wallet.allowed_transaction_types[]	Subset of transaction types permitted for this wallet
wallet.connected_external[]	External payment method IDs (Stripe cards, Plaid accounts, crypto addresses)
wallet.activity_feed_id	Linked activity feed for real-time transaction notifications
wallet.metadata	Full ComponentMetadata

3.4 Auto-Split Rules

A powerful Wallet feature is configurable auto-split: when income arrives in a wallet, it is automatically routed to multiple accounts according to defined percentage rules. This enables independent workers to automate their financial discipline without manual allocation:

Example: Jordan sets up the following auto-split on their Payroll Wallet: 60% → Personal Checking, 20% → Operations Account, 15% → Tax Reserve, 5% → Investment Account. Every time a contract payment or gig payout lands, kogi-bank automatically applies the split, and all four ledgers are updated atomically.

Auto-Split Rule Type	Description	Use Case
----------------------	-------------	----------

Percentage Split	Fixed % to each destination account	Tax reserve automation, savings allocation
Fixed Amount First	A fixed dollar amount to one account, remainder to another	Bill reserve before discretionary
Threshold-Triggered	Split activates only when deposit exceeds a threshold	Overflow to investment once bills covered
Category-Based	Route based on payer/source category	Client payments → Operations, marketplace → Marketplace Wallet
Goal-Linked	Route to account until a balance goal is reached, then switch	Fill tax reserve to \$X, then route to savings

3.5 External Payment Method Linking

Wallets can be linked to external financial instruments via the kogi-providers ProviderSystem:

- Bank accounts (ACH/wire) — linked via Plaid; balance visible, transfers initiated from kogi-bank
- Debit/credit cards — linked via Stripe; card-on-file for platform purchases
- PayPal / Venmo / Cash App — linked via OAuth; cross-platform transfers
- Crypto wallets — linked via WalletConnect; ETH, BTC, SOL, and ERC-20 tokens
- Robinhood / SoFi / brokerage accounts — read-only balance visibility; manual transfer
- Stripe Connect — seller payout routing for marketplace proceeds

4. LEDGER, JOURNALS & ACCOUNTING

4.1 Double-Entry Ledger

The kogi-bank ledger implements double-entry bookkeeping — the accounting standard used by every professional financial institution. Every financial event creates at least two ledger entries: a debit to one account and a credit to another. This ensures the ledger is always in balance, and that every transaction has a clear origin and destination.

The ledger is append-only and immutable. No entry is ever modified or deleted; corrections are made by posting a reversing entry. This design mirrors the Portfolio System's EventLog (also append-only) and provides a complete, tamper-evident audit trail for every financial event in the platform.

Accounting Principle Debits increase asset/expense accounts; decrease liability/equity/revenue accounts. Credits do the inverse. The fundamental equation $\text{Assets} = \text{Liabilities} + \text{Equity}$ holds at all times.

4.2 Ledger Entry Data Model

Field	Description
entry.id	Platform-unique UUID
entry.ledger_id	Parent ledger reference (one ledger per account owner)
entry.journal_id	Journal classification (income, expense, asset, liability, equity, escrow, campaign)
entry.transaction_id	Originating transaction UUID (groups the debit/credit pair)
entry.account_id	Account this entry affects
entry.type	debit credit
entry.amount	Decimal amount with currency code
entry.balance_after	Running balance of the account after this entry (computed, cached for performance)
entry.entry_type	income expense transfer escrow_lock escrow_release investment dividend fee tax refund donation grant payroll adjustment reversal
entry.description	Human-readable description
entry.reference	External reference (invoice ID, deal ID, order ID, campaign ID, etc.)
entry.counterparty_id	Entity on the other side of the transaction
entry.category	User-assigned or AI-auto-categorized expense/income category
entry.reconciled	Boolean: matched against external bank statement

entry.timestamp	Nanosecond-precision event timestamp
entry.metadata	Vector clock, policy_ids, tags, audit trail fields

4.3 Journal Classification

Every ledger entry belongs to a journal. Journals group entries by financial nature and drive the accounting reports, tax calculations, and P&L generation:

Journal	Entry Types	Drives
Income Journal	Client payments, marketplace proceeds, gig payouts, subscription revenue	Revenue recognition, P&L income line
Expense Journal	Subscriptions, tools, equipment, platform fees, professional services	P&L expense line, deductible tracking
Asset Journal	Equipment purchases, IP acquisitions, investment deployments	Balance sheet assets
Liability Journal	Loans, deferred revenue, outstanding payables	Balance sheet liabilities
Equity Journal	Equity issuances, cap table changes, vesting events	Balance sheet equity, cap table
Escrow Journal	Escrow locks and releases from marketplace/exchange deals	Deal accounting, held funds
Campaign Journal	Campaign contributions received and disbursed	Campaign accounting, donor records
Tax Journal	Tax reserve allocations, estimated tax payments, tax refunds	Tax liability tracking
Payroll Journal	Member/worker compensation payments	Payroll accounting, labor costs
Adjustment Journal	Corrections, reversals, write-offs	Error correction, reconciliation

4.4 Accounting Reports

kogi-bank generates the following standard accounting reports for individual workers and organizational entities, computed from the ledger in real-time:

Report	Description	Audience
Profit & Loss (P&L)	Revenue minus expenses over a defined period	Worker, Org, Cooperative
Balance Sheet	Assets, liabilities, and equity at a point in time	Org, Cooperative, Investor due diligence
Cash Flow Statement	Operating, investing, and financing cash flows	Worker, Org

Income Statement	Detailed income breakdown by client, project, and category	Worker, Org
Expense Report	Categorized expense breakdown; deductible flagging	Worker, Org
Invoicing Summary	Outstanding, paid, and overdue invoice status	Worker, Org
Tax Estimate	Self-employment tax liability estimate with quarterly breakdown	Worker
Payroll Summary	Compensation distributed to members/workers per period	Cooperative, Org
Campaign Report	Contributions, disbursements, and campaign outcome financials	Org, Campaign owner
Equity Register	Cap table, vesting schedules, ownership percentages	Org, Cooperative
Portfolio Financial KPIs	Revenue, burn rate, budget utilization, ROI per portfolio project	Worker, Org

4.5 Tax Management

kogi-bank provides autonomous tax management for independent workers and organizational entities:

- Self-employment tax estimation — automated quarterly SE tax calculation based on net income
- Deduction tracking — AI-auto-categorized expenses flagged as potentially deductible (home office, equipment, professional services)
- Tax reserve automation — configurable % auto-split from every income deposit into Tax Reserve Account
- Quarterly payment reminders — Oba agent notifies when estimated tax payments are due with amount and payment options
- 1099/W-9 management — request, collect, and store contractor tax documents for platform payouts above thresholds
- Multi-entity tax accounts — separate tax reserves for each organizational entity; consolidated reporting for affiliated entities
- International tax awareness — multi-currency income flagging; withholding tax alerts for cross-border transactions

5. TRANSACTION SYSTEM

5.1 Transaction Data Model

A Transaction is the atomic financial event that drives one or more ledger entries. Every movement of value between accounts — payment, transfer, escrow lock, payout, donation, investment — is represented as a Transaction.

Field	Description
tx.id	Platform-unique UUID
tx.type	payment transfer escrow_lock escrow_release invoice_payment payout donation investment refund fee tax_payment payroll campaign_contribution campaign_disbursement exchange_settlement swap adjustment
tx.status	initiated pending processing completed failed reversed disputed
tx.amount	{ value, currency, fx_rate_at_execution (for cross-currency) }
tx.from_wallet_id	Source wallet
tx.from_account_id	Source account (resolved from wallet)
tx.to_wallet_id	Destination wallet
tx.to_account_id	Destination account
tx.reference_type	invoice deal order campaign escrow payroll gig contract manual
tx.reference_id	ID of the linked reference entity
tx.description	Transaction description
tx.initiated_by	User or system agent that triggered the transaction
tx.approval_status	not_required pending_approval approved rejected
tx.approval_workflow_id	Linked ApprovalWorkflow ID (for multi-sig or policy-gated transactions)
tx.ledger_entries[]	Created debit/credit pair entry IDs
tx.fees[]	Platform or network fees applied to this transaction
tx.external_id	External payment processor reference (Stripe charge ID, ACH trace number, etc.)
tx.metadata	Full ComponentMetadata with audit trail

5.2 Transaction Types

5.2.1 Inbound Transactions

Type	Description	Trigger
Client Payment	Direct payment from a client for services rendered	Invoice marked paid; Stripe webhook
Marketplace Payout	Proceeds from a completed marketplace listing sale	Deal completion → escrow release
Gig Payout	Payment received for completed gig or contract milestone	Milestone approved in kogi-exchange
Campaign Contribution	Backer contributions to an active campaign	Campaign backing action
Investment Inflow	Capital received for equity or revenue share offering	Investment round closing
Donation Received	Non-equity charitable contribution	ActionKind::Donate in Portfolio System
Refund Received	Return of previously paid funds	Refund initiated by counterparty
Transfer In	Internal platform transfer from another account/wallet	Manual or auto-split transfer

5.2.2 Outbound Transactions

Type	Description	Trigger
Expense Payment	Payment for tools, subscriptions, services	Manual or auto-pay rule
Payroll Distribution	Compensation to team members or cooperative members	Payroll schedule trigger
Marketplace Purchase	Payment for a listing, asset, or work engagement	Deal accepted; escrow funded
Investment Deployment	Capital deployed into an equity round or campaign	Investment action approved
Donation Given	Charitable or mutual-aid contribution	ActionKind::Donate
Tax Payment	Estimated tax payment to tax authority	Manual or Oba-assisted
Platform Fee	kogi-bank / platform service charges	Transaction-triggered, automated
Escrow Funding	Locking funds into an escrow for an active deal	Deal accepted in marketplace
Transfer Out	Internal transfer to another account/wallet	Manual or auto-split

5.3 Payment Rails

kogi-bank supports multiple payment rails for inbound and outbound transactions, managed through the kogi-providers ProviderSystem:

Rail	Direction	Typical Latency	Use Case
Stripe	In + Out	Instant (card) / 2 days (payout)	Client card payments, marketplace payouts
ACH (via Plaid)	In + Out	1–3 business days	Bank-to-bank transfers, payroll
Wire Transfer	In + Out	Same / next day	Large international payments
PayPal	In + Out	Instant (platform) / 1-3 days (bank)	Small payments, contractor payouts
Ethereum / ERC-20	In + Out	Minutes (on-chain)	Token payments, crypto payouts, DeFi
Coinbase Commerce	In + Out	Minutes	Crypto invoice acceptance
Internal (kogi-bank)	In + Out	Instant	Platform-to-platform transfers
SEPA	In + Out	1 business day	European bank transfers

6. ESCROW SYSTEM

6.1 Concept

The kogi-bank Escrow System provides secure, conditional custody of funds for Marketplace and Exchange deals. Escrow ensures that buyers' funds are protected until agreed deliverables are confirmed, while sellers are guaranteed payment upon successful delivery — eliminating counterparty risk from platform transactions.

Every escrow in kogi-bank is a first-class financial entity with its own account (Escrow Reserve Account), ledger trail, governance policies, and milestone schedule. Escrows can be simple (single-payment on delivery) or structured (multi-milestone with partial releases).

6.2 Escrow Data Model

Field	Description
escrow.id	Platform-unique UUID
escrow.deal_id	Linked Deal ID from kogi-marketplace or kogi-exchange
escrow.type	simple milestone time_locked multi_party conditional
escrow.status	unfunded funded partial_release fully_released disputed refunded expired
escrow.buyer_id	Funding party UUID
escrow.seller_id	Receiving party UUID
escrow.total_amount	Total escrow value with currency
escrow.funded_amount	Amount currently held in escrow
escrow.released_amount	Cumulative amount released to seller
escrow.reserve_account_id	Escrow Reserve Account holding the funds
escrow.milestones[]	Array of EscrowMilestone entities
escrow.release_conditions	JSONB: conditions that trigger partial or full release
escrow.auto_release_after_days	If > 0: auto-releases N days after delivery confirmation if no dispute
escrow.dispute_window_days	Window after delivery in which buyer can raise a dispute
escrow.dispute_id	Linked Dispute ID if a dispute is in progress
escrow.metadata	Full ComponentMetadata with audit trail

6.3 Escrow Milestone Model

Field	Description
milestone.id	UUID
milestone.escrow_id	Parent escrow reference
milestone.name	Milestone name (e.g. 'Design Phase', 'Alpha Delivery')
milestone.amount	Amount to release upon this milestone's approval
milestone.due_date	Expected delivery date
milestone.status	pending delivered approved rejected disputed
milestone.deliverable_description	What must be delivered to trigger this milestone
milestone.delivery_evidence[]	File or link attachments submitted as delivery proof
milestone.approved_by	User ID of approver
milestone.approved_at	Approval timestamp
milestone.release_tx_id	Transaction ID of the corresponding escrow release payment

6.4 Escrow Lifecycle

Stage	Description	Ledger Effect
Deal Accepted	Buyer and seller agree on terms. Escrow record created.	No ledger entry yet
Funded	Buyer transfers funds to Escrow Reserve Account.	Debit: Buyer Checking. Credit: Escrow Reserve.
In Progress	Work underway. Funds locked.	No movement. Balance visible to both parties.
Milestone Delivered	Seller submits deliverable. Milestone status → 'delivered'.	No movement. Dispute window opens.
Milestone Approved	Buyer approves. Partial release triggered.	Debit: Escrow Reserve. Credit: Seller Payroll/Checking.
Dispute Raised	Buyer raises dispute within dispute window. Funds frozen.	No movement. Dispute entity created.
Dispute Resolved	Resolution reached via platform mediation or arbitration.	Release or refund per resolution.
Auto-Release	Buyer silent past dispute window. Auto-release fires.	Debit: Escrow Reserve. Credit: Seller.
Full Release	All milestones approved. Escrow closed.	Final debit/credit. Escrow account balance → 0.

Refunded	Deal cancelled before delivery. Buyer refunded.	Debit: Escrow Reserve. Credit: Buyer Checking.
----------	---	--

6.5 Dispute Resolution

When a buyer raises a dispute, kogi-bank freezes the affected escrow funds and creates a Dispute entity. The resolution process:

1. Dispute Raised — buyer submits dispute with reason, evidence, and requested remedy
2. Seller Notified — seller receives dispute notification and has a response window
3. Evidence Exchange — both parties submit evidence via Deal Room
4. Oba AI Summary — AI agent summarizes both positions and evidence for review
5. Mediation — platform mediation team reviews and attempts settlement (automated for small amounts)
6. Arbitration — for unresolved disputes above a threshold; third-party arbitration panel
7. Resolution — funds split, fully released, or fully refunded per resolution decision
8. Appeal — either party can appeal within N days; final decision is binding

7. INVESTMENT & FINANCIAL INSTRUMENTS

7.1 Concept

kogi-bank manages the full lifecycle of investment activity on the Kogi platform — from individual workers making equity investments in community campaigns, to cooperative members managing shared funds, to organizations issuing equity rounds. All investment activity is recorded as Portfolio Items, giving investors full portfolio-level visibility into their financial positions alongside their work and asset portfolios.

7.2 Investment Account & Position Model

Field	Description
investment.id	UUID
investment.investor_id	Investing entity UUID (user, org, collective)
investment.subject_id	The Portfolio Component, Campaign, or Financial Instrument being invested in
investment.subject_type	campaign equity_round revenue_share portfolio_asset external_instrument
investment.instrument_type	equity convertible_note revenue_share community_bond donation grant token portfolio_index
investment.amount	Capital committed: { value, currency }
investment.units	Equity units, tokens, or shares received
investment.unit_price	Price per unit at investment time
investment.status	committed funded active vesting matured exited written_off
investment.terms	JSONB: vesting schedule, cliff, liquidation preference, pro-rata rights, interest rate, maturity date
investment.portfolio_component_id	The investment is itself a Portfolio Item (type=Asset)
investment.return_to_date	Current computed return on investment
investment.distributions[]	Dividend or revenue share payments received
investment.metadata	Full ComponentMetadata

7.3 Investment Types

Instrument	Description	Settlement	Portfolio Integration
------------	-------------	------------	-----------------------

Equity Stake	Fractional ownership in a project, program, or organization	Cap table update in kogi-organizations; equity units in Equity Account	Investment item linked to Organization Portfolio Component
Revenue Share Note	Right to a % of future revenue from a portfolio component	Automated periodic transfer from Income Journal when revenue recognized	Revenue trigger linked to Asset Portfolio Component income
Convertible Note	Debt convertible to equity upon trigger event (e.g., next funding round)	Conversion event updates Equity Account; debt entry reversed	Liability Journal until conversion; Equity Journal after
Community Bond	Fixed-term debt issued by a cooperative or collective	Periodic interest payments + principal return	Liability on issuer's Balance Sheet; Asset on holder's
Donation / Grant	Non-repayable capital contribution	Single inbound transaction; no obligation created	Income Journal for recipient; Expense Journal for donor
Token / Digital Asset	Platform-native or external blockchain token holding	On-chain transfers via crypto bridge	Crypto Account balance; market value tracked via price feed
Portfolio Index Unit	Fractional exposure to a curated group of portfolio assets	Composite settlement across constituent assets	Index position as Asset in Investment Account
Pre-sale / Pre-purchase	Early commitment to purchase an asset before launch	Escrow until delivery; converts to purchase on delivery	Escrow Journal until resolved

7.4 Cap Table Management

For organizational entities (organizations, cooperatives, DAOs), kogi-bank maintains an integrated cap table within the organization's Equity Account and equity-linked Portfolio Components:

- Shareholder register — each equity holder's name, units held, % ownership, and class
- Vesting schedule tracking — cliff dates, vesting milestones, forfeiture events
- Dilution modeling — preview impact of new issuances on existing holders' %
- Liquidation waterfall — priority ordering for distributions and exit proceeds
- Transfer restrictions — governance-enforced restrictions on equity transfers
- Distribution history — dividend, revenue share, and buyback events per shareholder

7.5 Portable Benefits

kogi-bank manages portable benefits for independent workers — financial benefits that travel with the worker regardless of which clients or projects they work on:

Benefit	Account Type	Management
Health Insurance Reserve	Savings Account (tagged: health)	Monthly auto-allocation from income; provider payment automation
Retirement Savings	Retirement Account	Configurable contribution rate; linked to external IRA/401k providers
Emergency Fund	Savings Account (tagged: emergency)	Goal-based auto-save until target balance reached
PTO Reserve	Resource Account (time units)	Hours credited from completed contracts; withdrawal = paid leave period
Equipment / Tools Fund	Operations Account (tagged: equipment)	Depreciation tracking; replacement fund auto-save
Professional Development	Operations Account (tagged: education)	Budget for courses, conferences, certifications
Equity Holdings	Equity Account	Cap table positions from work contracts, cooperative membership, investments

8. INVOICING, BILLING & ORDERS

8.1 Invoice Data Model

Field	Description
invoice.id	UUID
invoice.type	client_invoice platform_bill subscription milestone_invoice retainer tax_invoice collective_invoice
invoice.issuer_id	Entity issuing the invoice (user, org)
invoice.recipient_id	Entity receiving the invoice
invoice.status	draft sent viewed overdue paid partially_paid disputed cancelled void
invoice.line_items[]	Array of { description, quantity, unit_price, subtotal, tax_rate, tax_amount }
invoice.subtotal	Pre-tax total
invoice.tax_amount	Computed tax (if applicable)
invoice.total	Final total due
invoice.currency	ISO 4217 currency code
invoice.due_date	Payment due date
invoice.payment_terms	net_30 net_15 due_on_receipt custom
invoice.payment_link	Stripe-hosted or platform-native payment URL
invoice.portfolio_component_id	Linked project, gig, or deal generating this invoice
invoice.contract_id	Linked contract reference if applicable
invoice.reminders[]	Scheduled reminder events and their sent timestamps
invoice.payments[]	Partial or full payment transaction IDs
invoice.metadata	Full ComponentMetadata

8.2 Invoice Lifecycle

State	Description	Trigger
Draft	Private; not yet sent	Manual creation or AI-drafted on deal completion
Sent	Delivered to recipient; payment link active	Send action or auto-send on milestone approval

Viewed	Recipient opened the invoice	Email/link open tracking
Overdue	Past due date with unpaid balance	Automated status transition at due date + grace period
Partially Paid	Partial payment received	Inbound payment < total
Paid	Full payment received; ledger updated	Final inbound payment transaction
Disputed	Recipient raised a payment dispute	Dispute action by recipient
Cancelled / Void	Invoice nullified before payment	Manual action; reversal entry posted

8.3 Subscription & Recurring Billing

For ongoing service relationships, kogi-bank supports automated recurring billing:

- Subscription plans — fixed recurring intervals (weekly, monthly, quarterly, annual)
- Retainer billing — regular invoices against an ongoing service retainer agreement
- Usage-based billing — invoices computed from tracked usage metrics (hours, units, compute)
- Auto-invoicing — invoice generated and sent automatically on schedule without manual action
- Failed payment retry — configurable retry logic for failed subscription payments (3/7/14 day retry schedule)
- Proration — mid-cycle plan changes automatically prorated on next invoice

8.4 Funding, Bids, Offers, Deals, Requests, Proposals & Contracts

kogi-bank is the financial settlement layer for all transactional entities that originate in kogi-marketplace and kogi-exchange. Each of these entity types has a corresponding financial lifecycle:

Entity	Financial Lifecycle	kogi-bank Role	Settlement
Bid	Buyer places bid; funds may be reserved as a bid bond	Optional bid bond hold in Escrow Reserve	Release on win; refund on loss
Offer	Seller quotes price; no immediate financial commitment	Price commitment recorded in Invoice Draft	Invoice generated on acceptance
Deal	Bid/offer matched or proposal accepted	Escrow funded by buyer; released to seller on completion	Multi-milestone escrow lifecycle
Request for Proposal	Buyer posts budget ceiling; no immediate commitment	Budget ceiling recorded; escrow provisioned on acceptance	Per accepted proposal terms

Proposal	Seller submits scope + price; no commitment until accepted	Proposal price drives Invoice Draft and Escrow plan	Invoice + Escrow on acceptance
Contract	Legally-binding engagement agreement	Milestone payment schedule drives automated invoicing	Per contract milestone schedule
Gig	Single deliverable work unit	Fixed-price escrow	Release on delivery approval
Task	Sub-unit of a contract or project	Tracked against parent contract budget	No separate escrow; covered by contract
Retainer	Ongoing access to worker time	Monthly recurring invoice	Auto-invoice + auto-payment from buyer wallet

9. CAMPAIGNS & GROUP ECONOMICS

9.1 Concept

Campaigns are the group economic engine of kogi-bank — enabling independent workers, organizations, collectives, and cooperatives to collectively raise, pool, and deploy capital and resources toward shared goals. A Campaign is a time-bounded fundraising or resource-gathering effort with a defined goal, a set of backers/contributors, and a structured disbursement mechanism.

Every Campaign has a dedicated Campaign Account in kogi-bank that holds contributed funds during the campaign period. Upon success, funds are disbursed per the campaign's terms. Upon failure, funds are automatically returned to contributors — atomically, without manual intervention.

9.2 Campaign Types

Type	Description	Financial Instrument	Disbursement
Fundraise	Donation-based funding for a project, mission, or cause	Donation (non-equity, non-repayable)	Lump sum on goal met or deadline reached
Equity Crowdfunding	Fractional ownership stake in exchange for capital	Equity units issued to backers	Equity registered in Equity Account; capital to Operating Account
Revenue Share Pool	Backers pool capital; share in future revenue from a Portfolio Component	Revenue Share Note	Periodic distribution from Income Journal when revenue recognized
Resource Campaign	Collective gathering of non-monetary resources (compute, licenses, materials)	Resource units committed	Resource Account credited per contribution
Group Buy	Collective purchase of an asset or license at a negotiated bulk rate	Shared ownership or individual licenses	Payment released to vendor on threshold reached
Community Bond	Debt issuance from a cooperative or collective to fund operations	Fixed-rate debt instrument	Principal + interest repaid per bond schedule
Pre-sale	Early commitment to purchase an asset before launch	Pre-purchase obligation	Escrow released to seller on delivery; refunded if not delivered
Mutual Aid Round	Community pool for member emergency support	Donation with soft reciprocity	Disbursed to requesting member per mutual aid policy

9.3 Campaign Data Model

Field	Description
campaign.id	UUID
campaign.type	fundraise equity revenue_share resource group_buy bond presale mutual_aid
campaign.owner_id	Campaign creator UUID (user, org, collective)
campaign.portfolio_component_id	Linked Portfolio Program or Project being funded
campaign.campaign_account_id	Dedicated Campaign Account holding contributed funds
campaign.goal	{ amount, currency } or { resource_units, unit_type }
campaign.raised	Current contributed total
campaign.backer_count	Number of unique contributors
campaign.backers[]	Array of { backer_id, amount, committed_at, status }
campaign.equity_terms	JSONB: % equity offered, pre-money valuation, class, rights, vesting (equity campaigns)
campaign.revenue_share_terms	JSONB: % of revenue, duration, payment frequency, portfolio component (revenue share campaigns)
campaign.bond_terms	JSONB: interest rate, maturity date, payment schedule (bond campaigns)
campaign.deadline	Campaign end date/time
campaign.all_or_nothing	Boolean: if true, contributions refunded if goal not met by deadline
campaign.status	draft active paused funded failed distributing closed
campaign.milestones[]	Use-of-funds milestones with release conditions
campaign.disbursement_tx_ids[]	Transaction IDs of disbursements to campaign beneficiaries
campaign.refund_tx_ids[]	Transaction IDs of contributor refunds (if campaign fails)
campaign.metadata	Full ComponentMetadata

9.4 Campaign Financial Lifecycle

Stage	Ledger Effect	System Action
Campaign Created	No ledger entry	Campaign Account provisioned; Portfolio Component linked

Contribution Received	Debit: Backer Wallet → Credit: Campaign Account	Backer record added; progress % updated; feed event published
Goal Reached	No movement yet	Success event published; milestone schedule activated
Milestone Released	Debit: Campaign Account → Credit: Owner Operating Account	Disbursement per milestone; backers notified
Equity Issued (Equity type)	Equity Account entry for backer; Cap table updated	Equity units minted and assigned to backer accounts
Campaign Failed / Deadline Missed	No movement yet	Failure event triggered; refund workflow initiated
Contributor Refunded	Debit: Campaign Account → Credit: Backer Wallet (×N)	Atomic multi-payout; all backers refunded simultaneously
Campaign Closed	Campaign Account balance → 0	Account archived; final report generated

10. COOPERATIVE, COLLECTIVE & ORGANIZATIONAL BANKING

10.1 Concept

kogi-bank provides full banking infrastructure for the non-individual entities of the Kogi platform: autonomous organizations (LLCs, Corps, Trusts, Funds), collectives (informal resource-sharing groups), cooperatives (member-owned economic entities), independent teams (squads, guilds, tribes), and federations (networks of linked entities).

Organizational banking in kogi-bank extends all the same account, wallet, ledger, and transaction primitives available to individual workers — but adds the organizational governance layer: multi-signatory accounts, member-split payroll, shared equity management, collective fund governance, and federation-level financial coordination.

10.2 Organizational Banking Setup

When an organizational entity is created in kogi-organizations, kogi-bank automatically provisions a standard set of accounts and wallets:

Entity Type	Auto-Provisioned Accounts	Auto-Provisioned Wallets	Governance Model
Autonomous Org (LLC, Corp, Trust)	Operating, Payroll, Reserve, Tax, Investment	Org Wallet, Payroll Wallet, Campaign Wallet	Single or multi-sig per bylaws; board approval for large transactions
Collective	Operating, Mutual Aid Fund, Resource Account	Collective Wallet, Campaign Wallet	Member vote for withdrawals above threshold; flat governance
Cooperative	Operating, Payroll (member distribution), Equity Pool, Reserve, Tax	Org Wallet, Payroll Wallet, Campaign Wallet	Democratic 1-member-1-vote governance; patronage distribution schedule
Independent Team	Project Account, Operating	Org Wallet	Team lead approval; optional multi-sig
Federation	Federation Reserve, Campaign Account	Collective Wallet	Inter-org approval workflow; federate-level governance

10.3 Cooperative & Collective Economics

10.3.1 Patronage Distribution

Cooperatives operate on a patronage (proportional to contribution) distribution model. kogi-bank computes and distributes patronage dividends from the cooperative's operating surplus:

- Revenue recognized in cooperative's Income Journal

10. Expenses deducted in Expense Journal; surplus computed
11. Governance vote approves distribution amount and timing
12. Patronage % computed per member based on their contribution (hours, revenue share, capital contribution)
13. Payroll Journal entries created for each member's distribution
14. Transactions executed from Operating Account to each member's Payroll Wallet

10.3.2 Resource Sharing Accounts

Collectives can operate Resource Accounts denominated in non-monetary units — compute hours, license seats, equipment time, or any custom resource unit. Members can contribute to and draw from these pools:

- Resource units contributed are credited to the Resource Account
- Resource units consumed are debited — tracked in Ledger with the same rigor as monetary transactions
- Resource pool governance policies (minimum balance, max single withdrawal, contribution schedules)
- Integration with kogi-exchange Resource Market for external resource acquisition and offering

10.3.3 Federation Financial Coordination

Federations — networks of linked organizations and collectives — can operate a Federation Reserve and federated campaign infrastructure:

- Federation Reserve — shared capital pool funded by member contributions; governed by federation policy
- Inter-federate transfers — member organizations can transfer resources between one another within the federation
- Federation Campaign — cross-organizational fundraising campaign where multiple entities contribute to a shared goal
- Consolidated reporting — federation-level financial overview aggregating all member entities' KPIs

11. AI INTEGRATION

11.1 Finance Agent (kogi-engine)

The Finance Agent is the AI sub-system dedicated to financial intelligence within kogi-bank. It runs on a continuous monitoring loop, consuming ledger events and portfolio financial signals to provide proactive insights, forecasts, and automated actions:

Capability	Description	Engine	Output
Income Forecasting	Predicts next 30/60/90 day income based on active contracts, seasonal patterns, and historical data	AnalyticsEngine + RecommendationEngine	Revenue projection with confidence intervals
Expense Analysis & Categorization	Auto-categorizes all transactions using semantic classification; identifies recurring and anomalous expenses	AnalyticsEngine	Categorized expense breakdown; anomaly alerts
Tax Estimation	Computes self-employment tax liability in real-time from YTD income and expense data	AnalyticsEngine	Quarterly tax estimate with breakdown
Cash Flow Forecasting	Projects account balances forward based on known upcoming transactions and forecasted income	OptimizationEngine	30-day cash flow projection; overdraft risk warning
Budget Utilization	Compares actual spend against budget allocations in Portfolio System; flags overruns	AnalyticsEngine + RiskEngine	Budget health score; overrun alerts
Savings Goal Tracking	Monitors progress toward defined financial goals; suggests auto-save adjustments	OptimizationEngine	Goal completion ETA; savings rate recommendation
Anomaly Detection	Flags unusual transactions — unexpected large charges, duplicate payments, unauthorized activity	RiskEngine	Fraud alert; anomaly report
Investment Performance	Computes ROI, IRR, and portfolio-level return	AnalyticsEngine	Investment performance dashboard

	on all investment positions		
Campaign Velocity Monitoring	Tracks campaign contribution rate vs. goal; predicts success probability and time-to-goal	AnalyticsEngine	Campaign health score; outreach suggestions
Payment Risk Scoring	Scores counterparties in marketplace deals by payment history and risk signals	RiskEngine	Counterparty risk score; escrow recommendation

11.2 Oba AI Agent Workflows for kogi-bank

11.2.1 Invoice Generation & Recovery

When a deal or gig milestone is completed in kogi-marketplace, Oba automatically drafts the corresponding invoice, attaches project context, and prompts the user for one-click send. For overdue invoices, Oba drafts escalating follow-up messages calibrated to the client relationship history and flags cash flow impact.

11.2.2 Morning Financial Briefing

Each morning, the Oba portfolio briefing includes a financial section: account balances at a glance, invoices due today, upcoming scheduled payments, current cash flow trajectory vs. quarterly OKR, tax reserve status, and any anomalies detected overnight — all as an AI-composed prioritized action list.

11.2.3 Tax Reserve Automation

After each significant income deposit, Oba notifies the user with a tax estimate for the deposit and confirms the auto-split to the Tax Reserve Account. At each quarter-end, Oba drafts the estimated tax payment, shows the exact amount, and offers one-click initiation via the user's linked bank or Stripe.

11.2.4 Payroll Distribution Assistance

For organizational and cooperative entities, Oba computes the payroll distribution schedule based on member contributions, patronage rules, and available operating surplus. It drafts the full payout transaction set for owner or governance review, and executes the batch payout atomically upon approval.

11.2.5 Budget Overrun Intervention

When a portfolio project's actual spend approaches or exceeds its budgeted ResourceAllocation (Budget kind), Oba surfaces the overrun alert with specific line items, computes the projected

overrun at current burn rate, and suggests specific interventions: scope reduction, timeline extension, or supplemental budget request.

11.2.6 Campaign Intelligence & Outreach

For active fundraising campaigns, Oba monitors contribution velocity and predicts the probability of reaching the goal by the deadline. When velocity drops below the needed rate, Oba identifies high-probability backers from the social graph and community data, drafts personalized outreach messages, and prepares them for owner approval and dispatch.

12. DATA ARCHITECTURE

12.1 PostgreSQL Schema

12.1.1 Core Tables

Table	Description	Key Columns
accounts	All account records for users and organizations	id, owner_id, owner_type, type, denomination, currency, status, policy_ids JSONB, metadata JSONB
wallets	Wallet records linking owners to accounts	id, owner_id, type, linked_accounts JSONB, transaction_rules JSONB, auto_split JSONB, status
ledger_entries	Immutable double-entry ledger — append only	id, ledger_id, journal_id, transaction_id, account_id, type (debit/credit), amount, currency, balance_after, entry_type, reference_id, timestamp
transactions	Transaction records grouping ledger entry pairs	id, type, status, amount, currency, from_wallet_id, to_wallet_id, from_account_id, to_account_id, reference_type, reference_id, approval_status, fees JSONB, metadata JSONB
invoices	Invoice records	id, type, issuer_id, recipient_id, status, line_items JSONB, total, currency, due_date, payment_terms, contract_id, metadata JSONB
escrow_accounts	Escrow record per deal	id, deal_id, type, status, buyer_id, seller_id, total_amount, funded_amount, released_amount, milestones JSONB, release_conditions JSONB, dispute_id
campaigns	Campaign records	id, type, owner_id, portfolio_component_id, campaign_account_id, goal JSONB, raised, deadline, status, backers JSONB, equity_terms JSONB, milestones JSONB
investments	Investment position records	id, investor_id, subject_id, instrument_type, amount, units, status, terms JSONB, distributions JSONB, metadata JSONB

disputes	Dispute records for escrow and payment disputes	id, type, initiator_id, respondent_id, escrow_id, status, evidence JSONB, resolution JSONB, timeline JSONB
tax_estimates	Computed tax estimate records	id, owner_id, period, gross_income, deductions, net_income, estimated_tax, breakdown JSONB, computed_at
external_payment_methods	Linked external accounts and cards	id, owner_id, provider, method_type, external_id, metadata JSONB, status

12.2 Kafka Event Topics

Topic	Events	Consumers	Purpose
bank.transactions	transaction.initiated, transaction.completed, transaction.failed, transaction.reversed	AnalyticsEngine, kogi-portfolio, kogi-exchange, Feed Service	Real-time transaction stream for analytics and cross-module updates
bank.ledger	entry.posted	AnalyticsEngine (cashFlow signal), Tax pipeline	Append-only ledger stream for downstream analytics
bank.invoices	invoice.created, invoice.sent, invoice.paid, invoice.overdue, invoice.disputed	AnalyticsEngine, Oba Agent, Feed Service	Invoice lifecycle events; trigger reminders and briefing cards
bank.escrow	escrow.funded, escrow.milestone_released, escrow.disputed, escrow.refunded, escrow.closed	kogi-marketplace, kogi-exchange, AnalyticsEngine	Escrow lifecycle; trigger deal status updates
bank.campaigns	campaign.created, campaign.backed, campaign.funded, campaign.failed, campaign.distributed	kogi-marketplace, kogi-organizations, AnalyticsEngine	Campaign lifecycle; trigger equity issuance, refunds, notifications
bank.investments	investment.committed, investment.funded, investment.distribution_paid, investment.exited	kogi-portfolio, kogi-organizations, AnalyticsEngine	Investment lifecycle; cap table and portfolio updates

12.3 Redis Caching

- Account balances — hot-path cached per account ID; TTL 60s; invalidated on every new ledger entry
- Wallet aggregated balance display — computed aggregate for UI; TTL 30s

- Active escrow state — deal-in-progress escrow status for Deal Room widget; TTL 10s
- Invoice summary cache — outstanding invoices per user; TTL 120s; invalidated on payment
- Campaign progress — contribution totals and backer counts; TTL 30s
- Tax estimate cache — current-period estimate; TTL 1 hour; invalidated on new income entry

13. UX ARCHITECTURE & USER FLOWS

13.1 kogi-bank UI Structure

View	Description
Bank Home / Dashboard	Account balances at a glance, recent transactions, upcoming invoices, AI financial summary, quick actions
Accounts	All accounts with type labels, balances, status; create account, link external, view ledger
Wallets	Wallet grid with balance display, linked accounts, auto-split rules; add wallet, configure rules
Ledger	Paginated, filterable ledger view with journal classification; search by date, type, reference, amount
Transactions	Transaction history with status, type, counterparty; filter by wallet, account, date, type
Invoices	Invoice management: create, send, track, collect; overdue alerts; AI draft button
Escrow	Active escrows with milestone tracker, release controls, dispute button; deal context panel
Investments	Investment portfolio: positions, returns, distributions; campaign tracking, equity register
Campaigns	Campaign management: create, track, disburse; backer list; contribution analytics
Reports	P&L, Balance Sheet, Cash Flow, Tax Estimate, Expense Report — downloadable and shareable
Tax Center	YTD income, estimated tax liability, deduction tracker, quarterly reminders, 1099 management
Org Banking	Multi-entity banking: org account overview, payroll, equity register, governance approvals
Settings	Payment methods, auto-split rules, notification preferences, compliance documents

13.2 Core User Flows

13.2.1 Independent Worker — Setup Financial Infrastructure

15. Open kogi-bank → Bank Home → Account Setup Wizard
16. Create Personal Checking, Operations, and Tax Reserve accounts
17. Configure auto-split: 60% Checking / 20% Operations / 20% Tax Reserve
18. Link external bank via Plaid and credit card via Stripe
19. Connect Payroll Wallet (receives client payments) to Personal Checking as primary

- 20. Set quarterly tax reminder — Oba schedules alerts
- 21. Financial infrastructure is now operational — every income deposit auto-routed

13.2.2 Create and Collect an Invoice

- 22. Complete work milestone → kogi-gig or kogi-marketplace triggers invoice draft in Oba
- 23. Oba pre-populates: client name, line items from project scope, amount, due date
- 24. User reviews draft → adjusts if needed → one-click Send
- 25. Client receives invoice with payment link (Stripe-hosted or platform payment)
- 26. On payment: transaction created → ledger entries posted → auto-split fires → notification sent
- 27. Invoice status → Paid; project financial KPIs updated in Portfolio System

13.2.3 Escrow — Hire a Worker

- 28. Accept Proposal in kogi-marketplace → Escrow created automatically
- 29. Fund Escrow: select Marketplace Wallet → Escrow Wallet → Confirm
- 30. Funds locked: Debit Marketplace Wallet account, Credit Escrow Reserve Account
- 31. Track milestones in Deal Room → escrow status widget shows locked amount
- 32. Seller delivers milestone → approve in Deal Room → milestone payment released
- 33. Final milestone approved → full escrow released → review workflow initiated

13.2.4 Cooperative — Monthly Payroll Distribution

- 34. Month-end: Oba triggers payroll computation for Cooperative entity
- 35. Oba computes member distributions based on hours contributed and patronage formula
- 36. Payroll summary presented to cooperative governance for approval vote
- 37. Vote passes → Oba prepares batch payout transaction set
- 38. Owner or multi-sig approvers confirm → batch executed atomically
- 39. All member Payroll Wallets credited simultaneously → Payroll Journal updated

13.2.5 Launch an Equity Campaign

- 40. Open kogi-bank → Campaigns → New Campaign → Type: Equity
- 41. Link to Portfolio Project/Program → set goal, % equity offered, valuation, deadline
- 42. Oba drafts campaign description from portfolio context → publish to kogi-marketplace
- 43. Backers contribute → funds held in Campaign Account
- 44. Goal reached → governance approval → equity units issued to backers' Equity Accounts
- 45. Capital disbursed from Campaign Account to org Operating Account per milestone
- 46. Cap table updated in kogi-organizations → investment positions recorded in kogi-bank

14. TECHNICAL ARCHITECTURE

14.1 Service Architecture

Service	Description	Technology
kogi-bank (Go)	Primary REST/gRPC API — account, wallet, transaction, and campaign management	Go, PostgreSQL, Kafka producer
kogi-ledger (Go)	Append-only ledger service — all write operations to ledger_entries table; balance computation	Go, PostgreSQL (SERIALIZABLE isolation for ledger writes)
kogi-escrow (Go/Rust)	Escrow lifecycle management; integrates with kogi-marketplace deal events	Go service + Rust escrow state machine
kogi-invoices (Go)	Invoice CRUD, send, payment tracking, recurring billing	Go, PostgreSQL, Stripe webhooks
kogi-campaigns (Go)	Campaign lifecycle, contribution tracking, disbursement, refund orchestration	Go, PostgreSQL, Kafka
kogi-tax (Go + Scala)	Tax estimation, deduction tracking, 1099 management	Go (API), Scala (data pipeline for YTD computation)
kogi-engine (Scala)	Finance Agent analytics, forecasting, anomaly detection, cashFlow signals	Scala 3, gRPC server
kogi-payments-bridge (Go)	External payment rail integration: Stripe, Plaid, ACH, crypto bridges	Go, Stripe SDK, Plaid SDK, Web3 providers

14.2 Critical Design: Ledger Consistency

The ledger is the most critical data store in kogi-bank. Its consistency guarantees are the foundation of all financial trust in the platform. The following design decisions enforce correctness:

- Append-only writes — no UPDATE or DELETE on ledger_entries; corrections via reversal entries only
- Serializable isolation — all ledger writes execute at SERIALIZABLE isolation level in PostgreSQL; no phantom reads
- Atomic debit/credit pairs — every transaction creates both entries in a single database transaction; partial pairs are impossible
- Idempotency keys — all transaction requests carry an idempotency key; duplicate submissions are safely detected and rejected
- Event sourcing — account balances are always recomputed from ledger history; never stored as a mutable balance field

- Saga pattern for distributed transactions — multi-service transactions (e.g., escrow release spanning kogi-bank + kogi-exchange + kogi-portfolio) use the Saga pattern with compensating transactions for rollback
- CRDT reconciliation — for multi-device and offline scenarios, CRDT-safe merge strategies align with the Portfolio System's existing CrdtLog

14.3 Technology Stack

Layer	Technology	Component	Notes
Core Services	Go	kogi-bank, kogi-ledger, kogi-escrow, kogi-invoices	REST + gRPC; Kafka producers
State Machine	Rust	Escrow state machine, BankSystem core module	Embedded in kogi-modules bank module
Intelligence	Scala 3 / JVM	kogi-engine Finance Agent	gRPC server; cashFlow signal feed
Database	PostgreSQL	Ledger, accounts, transactions, escrow, campaigns	SERIALIZABLE for ledger; standard for others
Cache	Redis	Balance cache, active escrow state, invoice summary	TTL-based invalidation on writes
Streaming	Apache Kafka	bank.transactions, bank.ledger, bank.escrow, bank.campaigns	All financial events streamed for analytics
Payment Rails	Stripe, Plaid, Coinbase	External payment bridge service	Managed by kogi-payments-bridge
Crypto	Ethereum / EVM, Solana	On-chain escrow (future), token bridge	Planned Phase 4 feature
Web Client	Angular + TypeScript	kogi-web-client bank module	WebSocket for real-time balance and escrow updates
Mobile	Kotlin / iOS	kogi-mobile-client bank module	Push notifications for transactions and escrow events

14.4 Security & Compliance

- PCI DSS compliance — all card data handled by Stripe; kogi-bank stores no raw card numbers
- All transactions require explicit user approval — no AI-initiated fund movements without user confirmation

- Multi-factor authentication required for transactions above configurable thresholds
- Encryption at rest (AES-256) for all financial data; TLS 1.3 in transit
- Audit trail via immutable ledger + Portfolio System EventLog — every action logged with actor, timestamp, and result
- Regulatory compliance hooks — KYC/AML screening for accounts above transaction thresholds (via third-party provider integration)
- GDPR / CCPA right-to-be-forgotten — personal financial data anonymization on account deletion (ledger entries preserved for regulatory purposes)
- Role-based access control — PermissionTier hierarchy governs all account and wallet operations

15. OPEN ITEMS & FUTURE CONSIDERATIONS

15.1 Near-Term Open Items

Item	Description
Ledger Persistence Optimization	Balance recomputation from full ledger history will become slow for long-lived accounts. Implement periodic balance snapshots with a verified snapshot + incremental replay approach for performance.
Saga Compensation Protocol	Distributed transactions spanning kogi-bank, kogi-exchange, and kogi-portfolio need a fully-specified Saga compensation map. Define compensating transactions for each step in the escrow, deal, and campaign settlement flows.
Crypto Bridge Specification	Token-based escrow and investment settlement requires a defined bridge protocol between kogi-bank and EVM / Solana chains. Specify the bridge adapter interface, key management, and on-chain escrow contract architecture.
KYC/AML Integration	Accounts above defined thresholds need Know Your Customer / Anti-Money Laundering screening. Select and integrate a third-party KYC/AML provider (e.g., Stripe Identity, Persona, Onfido) via kogi-providers.
Tax Engine Jurisdiction Handling	Tax estimation currently assumes US self-employment tax. Multi-jurisdiction tax logic (VAT, GST, withholding) needs a pluggable jurisdiction rules engine.
Cooperative Governance Voting	Payroll distribution and large withdrawal approval for cooperatives needs a lightweight on-platform voting mechanism. Define integration between kogi-organizations governance and kogi-bank multi-sig approval.
AI Finance Agent Writeback Protocol	Finance Agent produces forecasts and recommendations but needs a defined writeback protocol to create Invoice drafts, schedule transactions, and append financial governance notes — without bypassing PermissionTier checks.
Portable Benefits Provider Integration	Retirement, health, and insurance benefit accounts need integration with third-party benefit providers. Define the ProviderSystem integration pattern for benefit account management.

15.2 Phase 4 Considerations

- On-chain escrow smart contracts — for crypto/token deals; self-executing escrow without custodial counterparty risk
- Decentralized finance (DeFi) integration — yield on idle campaign and reserve accounts via DeFi protocols
- Autonomous financial agent — Oba acts on pre-approved rules autonomously: auto-pay recurring bills, auto-invest above a balance threshold, auto-rebalance tax reserve

- Federation-level financial clearing — cross-federation settlement and clearing for multi-platform (ShangoOS) economic activity
- Real-world asset (RWA) tokenization — platform infrastructure to tokenize and trade physical assets held by cooperative or org entities on-chain
- Automated payroll compliance — automated payroll tax filing, W-2/1099 generation, and employer of record functions for cooperative and org entities
- AI-driven investment recommendations — personalized investment opportunity matching based on portfolio risk profile, liquidity needs, and financial goals

© 2026 Kogi Platform — kogi-bank System Design Document — Confidential